



AI SOFTWARE DEVELOPMENT INTERNSHIP

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Week 5 — Detailed Task Sheet

Embeddings & Vector Databases • RAG Fundamentals Manual Retrieval-
Augmented Generation Pipeline • Git — `git revert`

PART C — RAG FUNDAMENTALS: BUILDING THE PIPELINE BY HAND

Practice Tasks

1. Add 3–4 short text documents (a couple of paragraphs each, any topic you like) using `add_document()`.

Outputs:

```
(venv) PS D:\Week5\Week5_PartC_RAG> python rag_pipeline.py

Source: rainforest.txt
Chunk: 0
Text:
    The Amazon rainforest is located mainly in South America.
    It is one of the most biologically diverse regions on Earth.

    The rainforest contains millions of plants, animals, and insects.
    It also plays an important role in the global carbon cycle and climate.

Source: python.txt
Chunk: 0
Text:
    Python is a popular programming language used in web development,
    automation, data analysis, artificial intelligence, and machine learning.

    Python uses simple and readable syntax. It supports important concepts
    such as variables, functions, loops, conditions, lists, and dictionaries.

○ (venv) PS D:\Week5\Week5_PartC_RAG>
```

2. Ask a question clearly answerable from one of the documents and confirm the retrieved chunks and the final answer both make sense.

Output:

```
• (venv) PS D:\Week5\Week5_PartC_RAG> python rag_pipeline.py
4 documents added successfully.
Question: Which planet is third from the Sun?

Retrieved Chunks:

Source: space.txt
Chunk: 0
Text:
    The Solar System contains the Sun and the objects that orbit it.
    The eight planets are Mercury, Venus, Earth, Mars, Jupiter, Saturn,
    Uranus, and Neptune.

    Earth is the third planet from the Sun. It has liquid water on its
    surface and supports a wide variety of life. The Moon is Earth's
    natural satellite and orbits around Earth.

Source: rainforest.txt
Chunk: 0
Text:
    The Amazon rainforest is located mainly in South America.
    It is one of the most biologically diverse regions on Earth.

    The rainforest contains millions of plants, animals, and insects.
    It also plays an important role in the global carbon cycle and climate.

Source: python.txt
Chunk: 0
Text:
    Python is a popular programming language used in web development,
    automation, data analysis, artificial intelligence, and machine learning.

    Python uses simple and readable syntax. It supports important concepts
    such as variables, functions, loops, conditions, lists, and dictionaries.

Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. Similarly, direct use of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream.

Final Answer:
Earth is the third planet from the Sun.
○ (venv) PS D:\Week5\Week5_PartC_RAG>
```

3. Ask a question that has nothing to do with any of your documents and confirm the model says it doesn't know, instead of making something up

Output:

```
Source: rainforest.txt
Chunk: 0
Text:
The Amazon rainforest is located mainly in South America.
It is one of the most biologically diverse regions on Earth.

The rainforest contains millions of plants, animals, and insects.
It also plays an important role in the global carbon cycle and climate.

Source: space.txt
Chunk: 0
Text:
The Solar System contains the Sun and the objects that orbit it.
The eight planets are Mercury, Venus, Earth, Mars, Jupiter, Saturn,
Uranus, and Neptune.

Earth is the third planet from the Sun. It has liquid water on its
surface and supports a wide variety of life. The Moon is Earth's
natural satellite and orbits around Earth.

Source: python.txt
Chunk: 0
Text:
Python is a popular programming language used in web development,
automation, data analysis, artificial intelligence, and machine learning.

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such as variables, functions, loops, conditions, lists, and dictionaries.

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Final Answer:
I don't have that information in the provided documents.
(vuvv) PS D:\Week5\Week5_PartC_RAG> python rag_pipeline.py
4 documents added successfully.
Question: What is the capital of Japan?
```

4. Print the source metadata for each retrieved chunk alongside the answer — this is your source attribution.

Output

```
Source: rainforest.txt
Chunk: 0
Text:
The Amazon rainforest is located mainly in South America.
It is one of the most biologically diverse regions on Earth.

The rainforest contains millions of plants, animals, and insects.
It also plays an important role in the global carbon cycle and climate.

Source: python.txt
Chunk: 0
Text:
Python is a popular programming language used in web development,
automation, data analysis, artificial intelligence, and machine learning.

Python uses simple and readable syntax. It supports important concepts
such as variables, functions, loops, conditions, lists, and dictionaries.

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Final Answer:
Earth is the third planet from the Sun.

Sources:
- space.txt (Chunk 0)
- rainforest.txt (Chunk 0)
- python.txt (Chunk 0)
(vuvv) PS D:\Week5\Week5_PartC_RAG>
```